PROJECT PRESENTATION

BLOOD DONOR MANAGEMENT SYSTEM

A Java + MySQL console-based application

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Problem Statement

Managing donor details and blood requests manually can make information difficult to organize and retrieve.

Scattered records

Donor names, blood groups and contact details may be stored in different places.

Slow searching

Finding donors by blood group and city can take time when records are not searchable.

Request tracking

Blood requests need a clear way to record details and track their status.

Project focus: improve record organization and retrieval—not replace medical judgement or blood-bank procedures.

Proposed Solution

A menu-driven Java application connected to a MySQL database.

01

Enter details

User selects an option in Command Prompt.

02

Process in Java

Main.java handles the selected operation.

03

Connect with JDBC

DBConnection.java opens the MySQL connection.

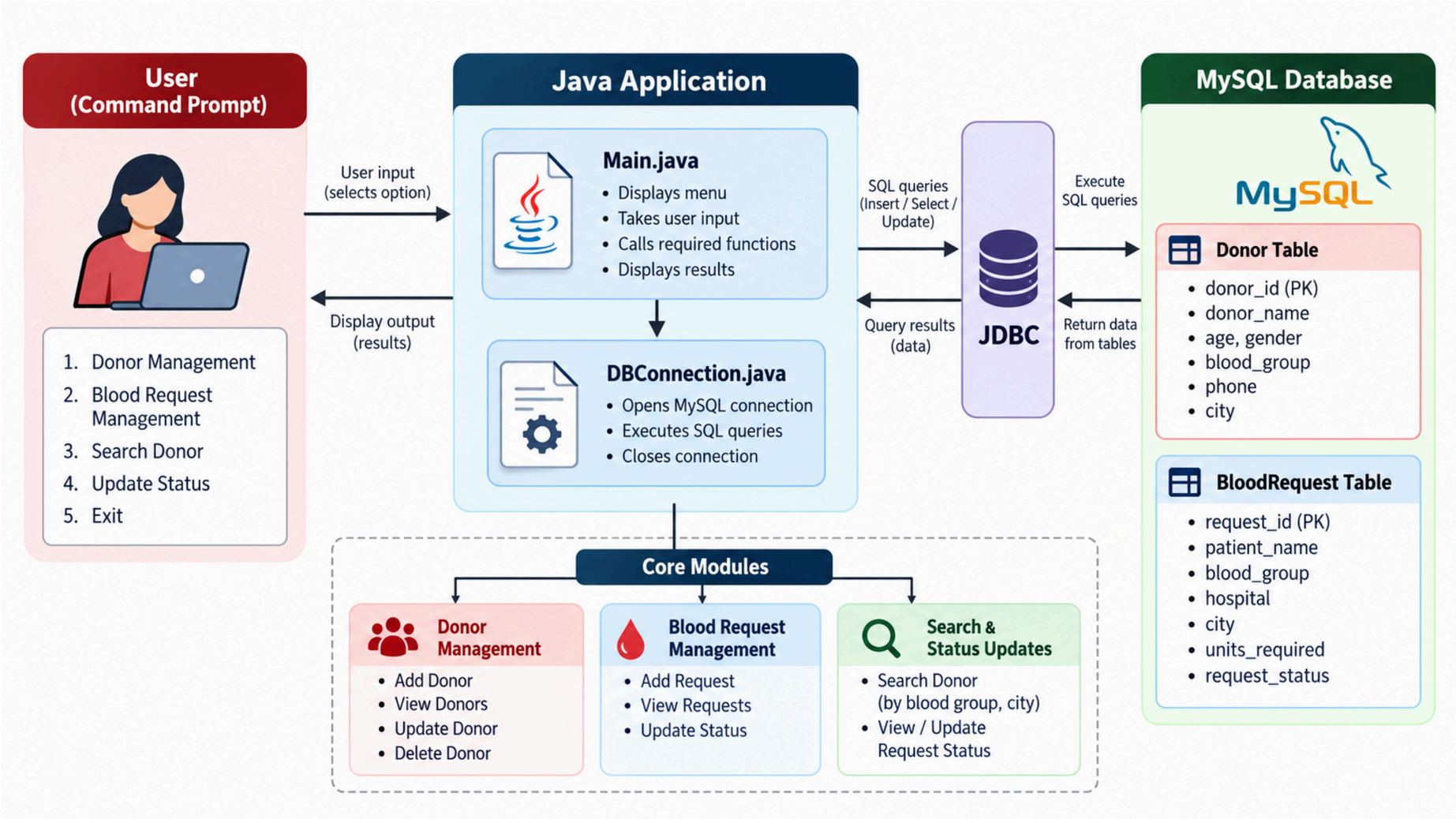
04

Store / retrieve

MySQL saves, updates, searches or returns records.

Core modules: Donor Management • Blood Request Management • Search & Status Updates

Architecture Diagram



Technology Stack

Java

Implements the menu, input handling and application logic.



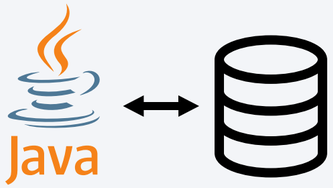
MySQL

Stores donor details and blood request records in tables.



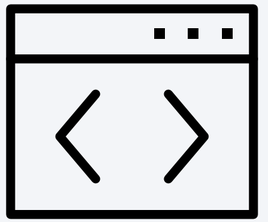
JDBC

Connects Java to MySQL and executes SQL statements.



Command Prompt + Workbench

Command Prompt runs the Java program; Workbench helps manage and inspect the database.



Key Java classes: Main.java | DBConnection.java

Java Concepts Used

The project uses core Java programming concepts along with JDBC for database connectivity.

Classes & Objects

Main and DBConnection are classes; Scanner, Connection and JDBC objects support operations.

Methods

Separate methods handle adding, viewing, searching and deleting donor records, and managing requests.

Control Statements & Loops

switch selects menu actions; if-else checks results; do-while repeats the menu; while reads result rows.

Exception Handling

try-catch handles SQL exceptions and displays database error messages.

User Input & Data Types

Scanner reads input; int stores numeric values and String stores names, blood groups, phone numbers and cities.

JDBC & PreparedStatement

JDBC connects Java to MySQL; PreparedStatement executes parameterized SQL queries.

Note: Inheritance, polymorphism and explicit abstraction are not directly demonstrated in the current code.

Database Design

Donor

donor_id — Primary key, auto-increment

donor_name — Name
age, gender — Personal details
blood_group — Blood group
phone — Contact number
city — City

BloodRequest

request_id — Primary key, auto-increment

patient_name — Patient name
blood_group — Required group
hospital, city — Request location
units_required — Units requested
request_status — Pending / Completed

Design note: primary keys uniquely identify each donor and each blood request.

Modules & Features

Add Donor

Save donor details to the database.

View Donors

Display all stored donor records.

Search Donors

Find records using blood group and city.

Add Blood Request

Store patient, hospital and units required.

View / Update Requests

Display requests and change their status.

Delete Donor

Remove a donor using the donor ID.

CRUD in this project: Create • Read • Update • Delete

Output & Testing

Replace the boxes below with screenshots from your own program.

```
C:\Users\HARSHITHA S\BloodDonorSystem>java -cp ".;lib\mysql-connector-j-26.7.0.jar" Main

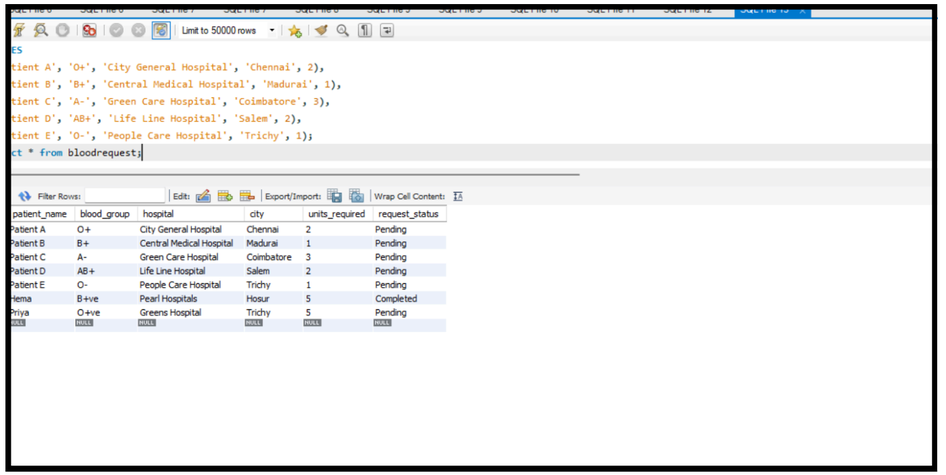
===== BLOOD DONOR MANAGEMENT SYSTEM =====
1. Add Donor
2. View All Donors
3. Search Donors
4. Add Blood Request
5. View Blood Requests
6. Update Blood Request Status
7. Delete Donor
8. Exit
```

```
Enter your choice: 1

--- Add Donor ---
Enter donor name: Rahul
Enter age: 25
Enter gender: M
Enter blood group: A+ve
Enter phone no: 98765432109
Enter city: Chennai
Donor details saved successfully!
```

```
Enter your choice: 4

--- Add Blood Request ---
Enter patient name: Priya
Enter blood group required: O+ve
Enter hospital name: Greens Hospital
Enter city: Trichy
Enter units required: 5
Blood request saved successfully!
```



```
ES
tient A', 'O+', 'City General Hospital', 'Chennai', 2),
tient B', 'B+', 'Central Medical Hospital', 'Madurai', 1),
tient C', 'A-', 'Green Care Hospital', 'Coimbatore', 3),
tient D', 'AB+', 'Life Line Hospital', 'Salem', 2),
tient E', 'O-', 'People Care Hospital', 'Trichy', 1);
ct * from bloodrequest;
```

patient_name	blood_group	hospital	city	units_required	request_status
Patient A	O+	City General Hospital	Chennai	2	Pending
Patient B	B+	Central Medical Hospital	Madurai	1	Pending
Patient C	A-	Green Care Hospital	Coimbatore	3	Pending
Patient D	AB+	Life Line Hospital	Salem	2	Pending
Patient E	O-	People Care Hospital	Trichy	1	Pending
Hema	B+ve	Pearl Hospitals	Hosur	5	Completed
Priya	O+ve	Greens Hospital	Trichy	5	Pending

Conclusion and Future Scope

CONCLUSION

The Blood Donor Management System simplifies donor record management and organizes blood request handling through a Java + MySQL database system. It helps in faster retrieval of donor information, better tracking of blood requests, and overall improved coordination.

“A simple step towards a bigger impact.”

FUTURE SCOPE

1. **Graphical User Interface: Replace the console with a user-friendly GUI for easier and intuitive interaction.**
2. **Donor Eligibility & Availability Tracking: Include health criteria and track donor availability for better matching.**
3. **Blood Bank Inventory Management: Maintain real-time stock levels of different blood groups.**
4. **Notifications: Send SMS or email alerts for new blood requests, status updates, and reminders.**
5. **Login & Role-Based Access Control: Add secure login with different roles for admin, staff, and users.**

Thank you

“Every drop counts. Every donor matters. Every life is precious.”

